An IoT-Enabled Hybrid Soft Sensor for Real-Time Residual-Moisture Monitoring and Resource Accounting in Compact Wool Scouring Machines

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***Abstract*—Wool scouring requires substantial water and electricity, while residual moisture is normally assessed off-line. This study presents an Internet of Things (IoT) architecture and a hybrid soft-sensor formulation for a compact five-bowl wool scouring machine with an improved ribbed-drum squeezing device. A geometry-based prior is derived from 24 historical observations in a 2^3 factorial experiment, and a proposed regularised correction uses process measurements and sparse oven-dry samples. Reanalysis of the historical data gave a four-coefficient moisture model that passed the reported Cochran and lack-of-fit criteria; its in-sample RMSE was 0.59 percentage points across 24 readings (0.23 points across eight run means). Historical factory records reported 65–70 % versus 55–60 % post-squeezing moisture, 59.6 versus 14.0 m^3 of water per shift, and 13 versus 4.2 kW installed power for the existing and improved machines, respectively. These results describe the mechanical process, not performance of the proposed IoT system. Sensor calibration, held-out soft-sensor accuracy, packet delivery, network latency, and metered per-kilogram resource use remain to be measured in a prospective field study. In a simulation under stated assumptions, the hybrid estimator kept a root-mean-square error of 0.42–0.50 percentage points after a change of squeezing geometry, versus 1.0–2.2 points for purely data-driven models; the server software processed a message in a median of 2.0 ms on a workstation. The paper therefore reports a reproducible retrospective baseline, a simulation and bench study, and an implementation and validation protocol, not a validated factory IoT deployment.**

***Index Terms*—Industrial Internet of Things, soft sensor, wool scouring, edge computing, resource efficiency, textile manufacturing, digital twin.**

I. Introduction

PRIMARY wool processing—opening, scouring, squeezing and drying—determines the quality and cost of all downstream textile products. Scouring removes grease, suint and mineral dirt in a sequence of bowls filled with hot alkaline liquor; between bowls, squeezing devices remove the dirty liquor from the wool. Insufficient squeezing carries dirt into the next bowl, shortens liquor life and increases water and detergent consumption, and every percentage point of residual moisture must later be evaporated in the dryer. Scouring is a major contributor to the water footprint and life-cycle burden of wool products [3], [4], and its highly polluted effluent requires costly treatment [5], [6].

In regions such as Uzbekistan, where wool comes largely from Karakul sheep and is processed in small, decentralised plants, large centralised scouring lines are uneconomic. A compact, resource-saving machine with an improved squeezing device (ribbed squeeze drum, guide slat drum and wrapped mesh surface) was developed and patented in previous work [1], [2]; factory tests reported a reduction of residual moisture from 65–70 % to 55–60 % and of installed power from 13 to 4.2 kW. However, such machines are operated without sensors: operators cannot see liquor temperature or dirtiness, moisture is measured only by occasional oven-dry samples, and resource consumption is not accounted per kilogram of wool.

Related work. Industrial IoT increasingly moves analytics to the edge of the network [7], [8], and IoT/big-data frameworks have been demonstrated for health-state monitoring, fault detection and cloud–edge–device collaborative monitoring of manufacturing systems [9]–[11]. Low-cost nodes make such monitoring affordable [12], [13], and digital retrofitting is the practical route for legacy machines that cannot be replaced [14]. Digital twins built from sensor data extend this to model-based operation [15]–[17]. In the textile sector, Industry 4.0 and AI work concentrates on spinning, inspection and garment production [18]–[21]; energy benchmarking of wet processing in SMEs has been proposed but relies on manual audits [22].

Soft sensors estimate hard-to-measure quality variables from routine measurements [23]. Purely data-driven models need many labelled samples, which are expensive when the reference is a laboratory analysis [24]; gray-box and physics-informed models reduce this need and improve explainability [25], [26], while adaptive schemes keep the model calibrated online [27]. In IoT settings, self-improving soft sensors have been used to reduce costly measurements [28]. To our knowledge, however, no work instruments the wool scouring/squeezing stage, and none uses a design-of-experiments model of the machine as the prior of an IoT soft sensor.

Contributions.

1) A low-cost (≈ US$ 200) IoT architecture for wool scouring machines with an ESP32 edge node, HTTP/MQTT

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transport, server-side analytics and an installable bilingual (EN/RU) web/mobile application.
2) A hybrid soft sensor for residual moisture that combines a 2^3 factorial regression model of the squeezing device (geometry prior) with an online, regularised correction for operating conditions, calibrated from sparse laboratory samples.
3) Real-time per-kilogram resource indicators (kWh/kg, L/kg) and rule-based fault alarms (heater failure, low level, dirty liquor, drum speed deviation).
4) Reanalysis of the historical mechanical-process baseline, a reproducible simulation and bench study, and an open-source IoT implementation (firmware, ready-made binaries, server, application) at https://github.com/nasin0764-bit/woolwash-iot; the prospective field validation is defined but has not yet been completed.

II. Machine and Squeezing Model

The machine comprises five bowls with stated internal dimensions of $2500 \times 600 \times 515$ mm [1][TODO: the source also states 2.5 m^3 per bowl, whereas these dimensions give 0.77 m^3; confirm from the engineering drawing]; moving rakes (length 250 mm, historically calculated linear speed 0.15 m/s [1], to be verified in the field) transport the wool, and each bowl is followed by a squeezing device with a ribbed drum of diameter D, a mesh of hole diameter d and a drum–mesh gap h [1].

A. DoE prior

A 2^3 full-factorial experiment with three replicates ($N = 8$, $m = 3$) with coded factors $X_1 = (d - 2)/1$, $X_2 = (D - 250)/50$, $X_3 = (h - 3)/1$ (mm) gave, after Student's test ($t_{0.05;16} = 2.12$), the adequate model ($F = 0.71 < F_{0.05;4;16} = 3.01$)

$$W_0 = 57.31 + 0.81X_1 - 0.95X_2 + 0.30X_3 \quad [\%]. \tag{1}$$

B. Hybrid soft sensor

Equation (1) captures geometry only. During operation, moisture also depends on the drum angular speed ω, the feed rate q, the liquor temperature T and the liquor turbidity τ. With normalised deviations $\boldsymbol{\phi} = [(\omega - \omega_0)/s_\omega, (q - q_0)/s_q, (T - T_0)/s_T, (\tau - \tau_0)/s_\tau]^\top$ we model

$$\hat{W} = W_0(d, D, h) + \mathbf{w}^\top \boldsymbol{\phi} + c, \tag{2}$$

where $(\mathbf{w}, c)$ minimise $\sum_i (W_i - W_{0,i} - \mathbf{w}^\top \boldsymbol{\phi}_i - c)^2 + \lambda \|\mathbf{w}\|^2 + \lambda_c c^2$ over laboratory samples W_i taken after the squeezer and dried at 105 °C. The model is refitted on the server each time a sample is added, in three stages: with $n = 0$ samples it is the DoE prior; for $1 \le n < 5$ only the plant offset is estimated, $c = \sum_i r_i/(n + \lambda_c)$ with $\mathbf{w} = \mathbf{0}$; for $n \ge 5$ the full correction is fitted ($\lambda = 1$). The offset penalty $\lambda_c = \sigma_{\text{lab}}^2/\tau_c^2$ corresponds to a Gaussian prior $c \sim \mathcal{N}(0, \tau_c^2)$, with the laboratory noise $\sigma_{\text{lab}} = 0.664$ pp taken from the pooled replicate variance of Table I and an *assumed* prior plant-offset uncertainty $\tau_c = 1$ pp. At a single, fixed geometry W_0 is a constant that the only weakly penalised bias c absorbs, so the hybrid estimator then nearly coincides with ordinary ridge regression; the prior adds information only at cold start, with few labels and when the geometry changes (another drum, gap or mesh, or another machine). Sec. VI quantifies both cases in simulation. The DoE regression is evaluated in Sec. V using its historical observations. A comparative assessment on temporally held-out field samples is reserved for Sec. VII.

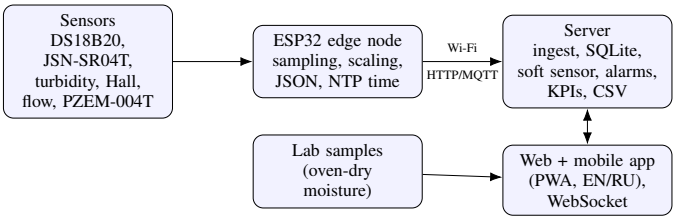


Fig. 1. Architecture of the proposed IoT framework.

III. IoT System Architecture

Fig. 1 shows the architecture. The node supports both HTTP and MQTT; the choice trades broker infrastructure against energy and overhead, as quantified in recent protocol comparisons [29], [30].

Proposed instrumentation and calibration. Five DS18B20 probes sit at mid-depth in the bowls, about 10 cm from the wall and away from steam injection; five JSN-SR04T rangefinders face the liquor from above; two turbidity probes sit in side-stream cups of bowls 1–2; a Hall pickup on the first squeeze-drum shaft gives ω; a pulse flowmeter measures fresh-water make-up; and a PZEM-004T interface meters electricity. An isolated 5-V supply and an IP65 enclosure are specified, mains work is done by an electrician, and the acquisition interval is 2 s. Wi-Fi and server settings are entered once in a captive set-up portal of the node, so one firmware image serves every machine. The image boots in Espressif's QEMU emulator up to Wi-Fi start (QEMU has no radio); this test exposed a crash of the default quad-I/O flash mode, so all builds use dual-I/O (DIO). Each channel will be calibrated against a reference (thermometer, depth gauge, 20-L volumetric collection, formazin standards, tachometer, revenue-grade energy meter); errors and uncertainties will be estimated from the paired readings, not taken from datasheets. These statements describe the design, not a completed installation.

IV. Experimental Setup

Retrospective material. The historical evidence is a 2^3 geometry experiment (8 settings, 3 moisture readings each) and mechanical-process comparisons at "Oqbo'tayeva Zulxumor" (Dehqonobod, Qashqadaryo) and "Hisori jun" (Oltinsoy, Surxondaryo), Uzbekistan [1]; it contains no IoT telemetry.

Prospective validation design (not yet performed). At least ten improved-process shifts (and five conventional ones where a comparable machine exists) with batch-wise wool type, feed mass, run hours and meter readings; oven-dry reference moisture after squeezing about every 30 min (105 °C, three replicates); at least 60 labelled observations for calibration and 20 on separate later days for testing, split by *day* to

TABLE I
HISTORICAL 2^3 FACTORIAL DATA FOR RESIDUAL MOISTURE W (%) [1]. LEVELS $-1/0/+1$: $d = 1/2/3$ MM, $D = 200/250/300$ MM, $h = 2/3/4$ MM

Run	X_1	X_2	X_3	Replicates	Mean ± SD	$\hat{W}$
1	−	−	−	57.0, 57.5, 58.3	57.60 ± 0.66	57.15
2	+	−	−	58.3, 58.5, 58.7	58.50 ± 0.20	58.78
3	−	+	−	54.5, 55.2, 55.4	55.03 ± 0.47	55.26
4	+	+	−	57.2, 56.2, 57.4	56.93 ± 0.64	56.88
5	−	−	+	58.5, 57.2, 56.8	57.50 ± 0.89	57.74
6	+	−	+	59.8, 59.5, 59.0	59.43 ± 0.40	59.37
7	−	+	+	55.1, 56.3, 56.2	55.87 ± 0.67	55.85
8	+	+	+	57.1, 58.8, 57.0	57.63 ± 1.01	57.48

prevent leakage; and per-packet sequence numbers, transmit/receive timestamps, RSSI and outages. The analysis is scripted (`tools/analyze_field.py`). These are targets, not completed sample counts.

V. RESULTS I: RETROSPECTIVE BASELINE FROM HISTORICAL DISSERTATION DATA

All values in this section are transcribed from the historical experimental program [1] (Tables 3.7, 3.9, 3.13 and Sec. 4.1 therein) and reanalysed; they are *not* measurements of the IoT system and serve only as the physical-process reference and as the prior of the soft sensor. Measurements of the IoT system are reported exclusively in Sec. VII.

A. Factorial experiment

The 2^3 design (Table I) comprised three residual-moisture measurements per treatment, giving 24 observations over eight geometry combinations. Unweighted least squares on the eight run means gives the main-effects model (1) ($b_0 = 57.3125$, $b_1 = 0.8125$, $b_2 = -0.9458$, $b_3 = 0.2958$); the interaction terms were not significant ($|t| \leq 1.08 < 2.12$). The within-run variances satisfy Cochran's homogeneity criterion ($G = 0.290 < 0.516$), and the model passes the lack-of-fit test ($F = 0.714 < F_{0.05;4,16} = 3.01$). These tests indicate no lack of fit at the eight measured corners but do not establish predictive accuracy outside the factor domain. The selected 2 mm/250 mm/3 mm geometry is an engineering compromise—narrower gaps caused wool jamming and damage, wider gaps reduced squeezing—and corresponds to a *predicted* residual moisture of 57.31 %; no separate center-point run was performed, and the minimum fitted corner (run 3, 55.26 %) is lower, so the setting is not the unconstrained statistical optimum.

B. Historical industrial observations

The improved design was tested at two enterprises in Uzbekistan (Qashqadaryo and Surxondaryo regions) [1]. Table II summarises the reported values and midpoint-based normalisations. Post-squeezing moisture decreased by about 10 percentage points and water use per shift by 76.5 %, but the improved line also had lower nominal throughput and lower clean-wool yield; the latter may reflect better impurity removal or fibre loss and cannot be resolved from aggregate values. Installed power is a nameplate value, not metered energy. The records lack matched batches, per-site variability and uncertainty, and are therefore descriptive evidence only—which is precisely the measurement gap the proposed IoT system addresses.

TABLE II
HISTORICAL COMPARISON REPORTED IN [1] (NOT IOT DATA)

Indicator	Existing	Improved	Change
Residual moisture (%)	65–70	55–60	−10 pp[a]
Water (m^3/shift)[b]	59.6	14.0	−76.5 %
Water reused (%)	70	80	+10 pp
Throughput (kg/h)[c]	700–800	400–450	−43.3 %[a]
Clean-wool yield (%)	64–66	60–62	−4 pp[a]
Installed power (kW)	13.0	4.2	−67.7 %
Water/throughput[d]	–	–	−58.5 %
Power/throughput[d]	–	–	−43.0 %
Scouring eff. (%)[e]	n/r	94.5; 92–95	–

[a] Range midpoints. [b] Units as in Table 3.13 of [1]; another table of the source uses litres per 450 kg. [c] Printed as "kg/s" in parts of the source; kg/h is the internally consistent reading. [d] Equal shift length and midpoint throughput assumed; nameplate power, not metered specific consumption. [e] Spring; autumn wool. n/r: not reported.

TABLE III
HISTORICAL IN-SAMPLE DOE FIT ONLY (NOT AN IOT OR HELD-OUT FIELD TEST)

Evaluation set	RMSE (pp)	MAE (pp)	R^2
Eight treatment means	0.229	0.185	0.969
All 24 historical readings	0.588	0.494	0.826

C. In-sample fit of the DoE prior

Table III reports residuals of model (1) *on the historical data from which its coefficients were estimated*. The 24-reading RMSE includes the variation among three repeats within each geometry. These values are not independent prediction errors; held-out accuracy is addressed only in Sec. VII.

D. Derived resource proxies

The only available water figures (59.6 and 14.0 m^3/shift) are historical reported process volumes with conflicting units elsewhere in the source. Under the additional assumptions of equal shift duration and midpoint throughputs of 750 and 425 kg/h, the ratio of water volume per kilogram for improved versus conventional processing is $(14/425)/(59.6/750) = 0.415$, or a 58.5 % lower *derived* water-to-throughput ratio. Actual L/kg cannot be established without the recorded shift duration and batch mass. Similarly, the nameplate-power/nominal-throughput proxies $13/750 = 0.0173$ and $4.2/425 = 0.00988$ kWh/kg assume full rated load and throughput; they are *not* metered specific energy. No IoT-versus-non-IoT savings or significance test is claimed.

VI. RESULTS II: SIMULATION AND BENCH STUDY

This section contains **no factory data**. It combines a simulation of the soft sensor under stated assumptions, a

TABLE IV
SIMULATED SOFT-SENSOR RMSE (PP), MEAN OVER 200 RUNS; NONLINEAR TRUTH, PLANT OFFSET 1.5 PP. TRANSFER: TRAINED ON 20 SAMPLES AT 2/250/3 MM, TESTED ON A NEW GEOMETRY

	Samples n, same geometry				Transfer	
Model	0	1	5	20	A	B
DoE prior only	2.09	2.09	2.09	2.09	2.09	2.09
Hybrid (proposed)	**2.09**	**1.23**	**0.82**	**0.50**	**0.50**	**0.50**
Hybrid, unstaged	2.09	2.09	0.85	0.50	0.50	0.50
Ridge, no prior	–	1.26	0.99	0.47	1.05	2.11
Random forest, no prior	–	1.26	0.91	0.64	1.13	2.15

A: D 250→300 mm. B: $d/D/h$ = 2/250/3 → 3/200/4 mm.

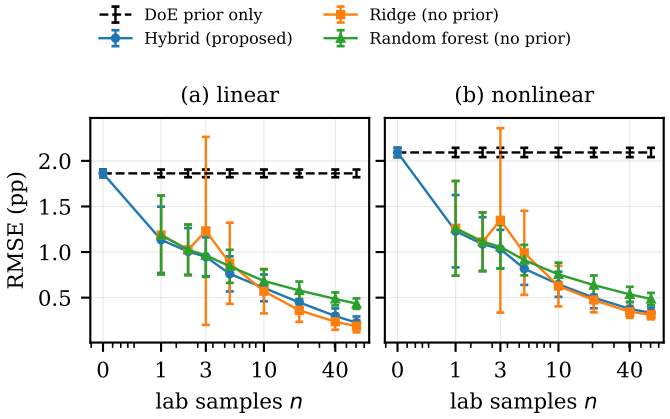


Fig. 2. Simulated learning curves (mean ± SD of RMSE over 200 runs, plant offset 1.5 pp) for (a) the linear and (b) the nonlinear ground truth. Simulation under assumed operating effects, not factory data.

measurement of the server software on a workstation, and a verification of the alarm logic on the simulator. All numbers are reproduced by `tools/sim_study.py` in the open repository.

A. Soft-sensor sample efficiency and geometry transfer (simulation)

Set-up. Ground-truth residual moisture is the DoE model (1) plus a plant offset (0 or 1.5 pp) and *assumed* operating effects of $+0.40$ pp per rad/s of ω, $+0.02$ pp per kg/h of q, -0.08 pp/°C and $+0.002$ pp/NTU, with $\omega \sim \mathcal{N}(16, 1)$ rad/s, $q \sim \mathcal{N}(425, 25)$ kg/h, $T \sim \mathcal{N}(55, 3)$ °C and $\tau \sim \mathcal{U}(200, 800)$ NTU. A *nonlinear* variant adds a quadratic drum-speed term, a feed–temperature interaction and a turbidity threshold effect that none of the models contains. Laboratory labels carry Gaussian noise with $\sigma_{\text{lab}} = 0.664$ pp (Table I). Each configuration is repeated 200 times; RMSE is computed against the noise-free truth on 300 test points.

Findings. (i) Purely data-driven models cannot predict before the first label, whereas the hybrid starts from the DoE prior. (ii) Shrinking the plant offset towards the prior gave the lowest *worst-case* error over the four scenarios at $n = 1$ and $n = 2$ (1.23 and 1.09 pp, versus 2.09 pp for the unstaged rule and 1.35 and 1.11 pp without shrinkage); with the linear truth at $n = 1$ it gave 1.01/1.13 pp (offset 0/1.5 pp) versus 0.79/1.86 pp unstaged, so it costs accuracy when the prior is already correct. At $n = 3$ the unstaged rule, which already fits the full correction, was slightly better (worst case 0.98 versus 1.03 pp). (iii) For $n = 5$ the proposed rule had the lowest error in all four scenarios (0.71–0.82 pp versus 0.73–0.85 pp unstaged and 0.85–0.99 pp for ridge). (iv) For $n \geq 20$ at fixed geometry, ordinary ridge regression was marginally better (e.g. 0.36 vs. 0.42 pp, linear truth), as expected from the equivalence noted in Sec. II-B; the random forest was worst. (v) After a geometry change the hybrid kept 0.42–0.50 pp, whereas the data-driven models rose to 1.0–2.2 pp, because they cannot represent geometry they were never trained on (Table IV, Fig. 2). These conclusions are conditional on the assumed effect sizes and must be confirmed with factory data (Sec. VII).

B. Server software benchmark (measurement on a workstation)

On an Intel Core i9-14900HX (Windows 11, Python 3.9), one firmware-format message is 638 bytes (27.6 MB/day per node at 2 s). Server-side processing (validation, soft-sensor prediction, alarm rules, SQLite insert) took a median of 2.0 ms (p95 2.4 ms, $n = 5000$). The HTTP round trip over the loopback interface, with a new TCP connection per message as in the firmware, had a median of 15.7 ms (p95 32.0 ms, $n = 2000$); a single sequential client achieved 62 messages/s, i.e. about 120 nodes at a 2-s period. These figures characterise the software only and exclude Wi-Fi, which is evaluated in Sec. VII.

C. Alarm-logic verification (simulator)

Faults were injected 20 times each into the simulator after a 1-h warm-up. All 60 were detected: heater failure in bowl 3 after a median of 92 s (max 110 s), a level leak in bowl 2 after 200 s (max 205 s), and a drum-speed drop in the first message after onset (≤1 s at the 2-s period). Both delays are set by the simulated thermal and level dynamics and by the thresholds, not measured plant behaviour. During 8 h (14 401 messages) of fault-free operation no fault alarm occurred. A plain 800 NTU threshold kept the *dirty-liquor advisory* active in 64 % of messages, because the simulated plant changes liquor only at 850 NTU. With a 30-s debounce and hysteresis (on 800, off 700 NTU) this fell to 37.8 %, and with per-machine commissioned limits (on 820, off 700 NTU) to 18.8 %; in both cases each of the 56 simulated liquor changes was preceded by an advisory, with 57 and 56 advisory episodes, respectively. The limits are therefore set per machine from the plant's liquor-change practice at commissioning (`PUT /api/limits`).

VII. RESULTS III: PROSPECTIVE FACTORY IOT VALIDATION

This section is reserved for measurements collected with the deployed IoT system at the factory. No value from Sec. V or Sec. VI is reused here, and simulator output is excluded. **Status: not yet performed**; the subsections define what will be reported. [TODO: Replace each subsection with measured results (n, mean ± SD, test days) once the field campaign is complete.]

Planned measures. (A) RMSE, MAE and R^2 of the hybrid, DoE-only, ridge and random-forest estimators on day-separated held-out oven-dry samples with identical test days for every model; the online prediction for each sample is stored before that sample is used for retraining. (B) Shift-level $\mathrm{kWh/kg} = \Delta E/M$ and $\mathrm{L/kg} = 1000\Delta V/M$ from metered electricity ΔE, water ΔV and weighed mass M. (C) Delivery ratio $100N_{\mathrm{received}}/N_{\mathrm{sent}}$ from per-packet sequence numbers and the median and 95th-percentile one-way delay $t_{\mathrm{receive}} - t_{\mathrm{transmit}}$ between NTP-synchronised clocks, with outages and RSSI. (D) Detection delay from an independently recorded onset of supervised faults (safety interlocks kept) to the first alert, and false alarms per hour of normal operation. None of these has yet been measured.

VIII. Discussion and Limitations

The historical data give a reproducible DoE baseline and an engineering rationale for the ribbed-drum and mesh configuration, but do not validate the IoT hardware, the estimator, the network or any savings from IoT feedback; the in-sample errors of Table III are no substitute for held-out accuracy, and the source's water volumes are inconsistent between tables. The improved machine's lower throughput and clean-wool yield may reflect better impurity removal or fibre loss; no matched fibre-quality assays settle this. Further limits are seasonal changes in greasy wool, sensor fouling in hot alkaline liquor, Wi-Fi attenuation in wet halls, a single site, and security controls (tokens, TLS, access roles, backups) that still have to be verified on the plant network. The simulation in Sec. VI depends on assumed operating-effect sizes; it shows where the DoE prior can help (cold start, few labels, geometry changes) and where it cannot (many labels at a fixed geometry), but not how large these effects are in a plant. Hence the present work is a system design, retrospective baseline and simulation study, not evidence of proven IoT effectiveness.

IX. Conclusion

An ESP32-based architecture and a DoE-informed hybrid soft-sensor formulation were specified for a compact wool-scouring machine. Reanalysis of 24 historical moisture observations confirmed an adequate four-term geometry model; the in-sample RMSE was 0.588 percentage points across individual readings. Historical records also suggest lower residual moisture and reduced water volume per shift for the improved *mechanical* process, subject to data comparability and unit limitations. A simulation with a staged, offset-shrinking calibration showed the benefit of the DoE prior to be confined to cold start, few labels and changes of geometry; the server software and alarm rules were verified on a workstation and simulator. No field measurements yet establish the accuracy, reliability, latency, alarm performance or per-kilogram resource benefits of the proposed IoT system. Independent, day-separated factory validation with calibrated sensors and metered consumption is required before such performance claims can be made.

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